SWEATERCO

GROUP 11 / TRACK 2 / SPRINT 1 REVIEW

# The Quick-Response Subcontracting Desk

*Who should make this batch? Answered in minutes, with the reasoning on record.*

THE NETWORK

*QuickStitch*
*SteadyHands*
*BudgetWorks*
*Little Loom*
*GiantWeave*
*Nimble Needle*
*OldMill*
*FreshStart*

*OldMill suspended. FreshStart on trial, max 300.*

PRIMARY OBJECTIVE **MINIMISE LATENESS**


Chen Zepeng (A0156784N) — Present (in person)
Duong Thanh Nhan (A0296326W) — Present (in person)
Joshua Wesley Gloor (A0296944J) — Present (in person)
Niyanthri Rajeswari Sridhar (A0351151U) — Present (in person)
Sundara Rajan Saadhvika (A0352594W) — Present (in person)
Tang Zheling (A0352177B) — Present (in person)


DSS5105 CAPSTONE / 16 OCTOBER 2026

# Today, one chat message is the whole decision

> *@Chen - order 042, 1,500 crewnecks. Send it to QuickStitch, need it back by the 19th.*
>
> HOW ALLOCATION IS DECIDED TODAY / COURSE BRIEF

Everything behind it lives in one dispatcher's head. Nobody can check it, and nothing is written down.

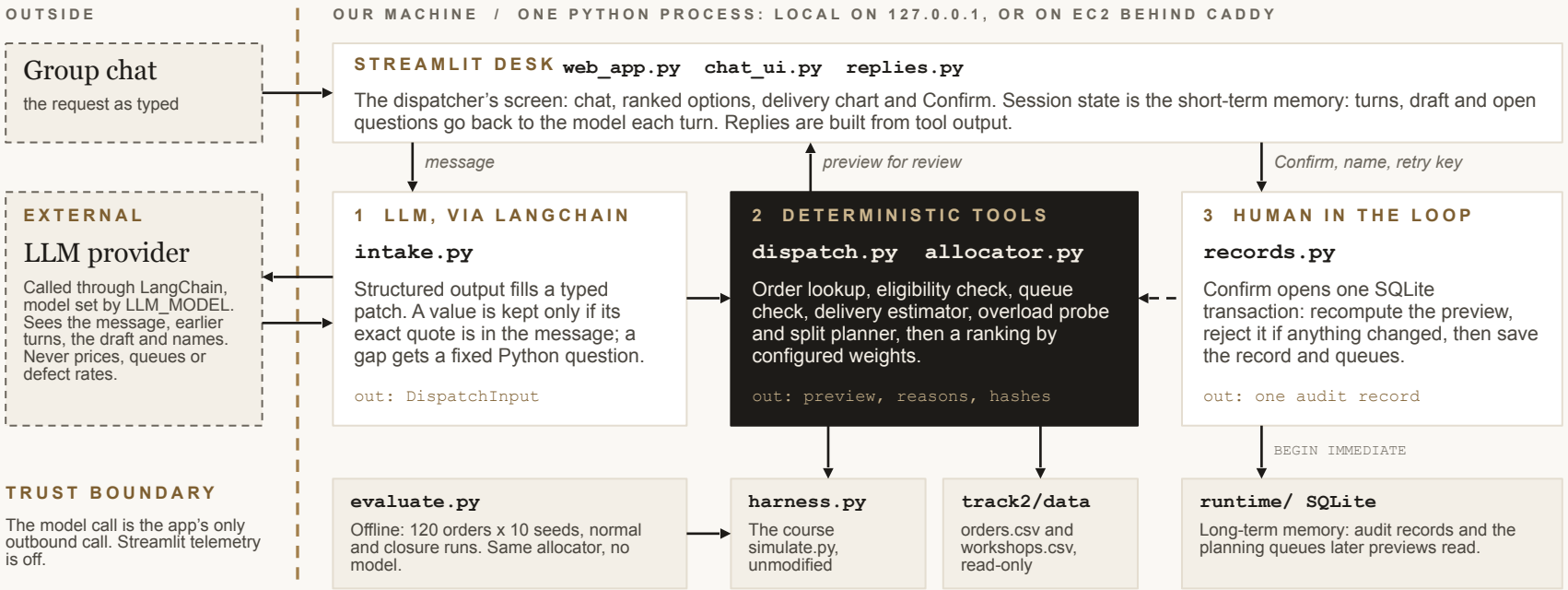
**7 of 30** requests in this morning's inbox are plain allocations. The other 23 need a question, a refusal, a constraint or bad news.

THREE QUESTIONS, EVERY TIME

1. **Can they make it?**
   Four of the eight make tops only, so they cannot take a scarf order.
2. **Can they take it now?**
   QuickStitch, "fastest in the region", already has four days of work queued.
3. **Are they allowed to?**
   OldMill is suspended after an audit. FreshStart is on trial, capped at 300 pieces.

*Get one wrong and the batch bounces back a day later. In a ten-day cycle, that is the margin gone.*

# The model reads. Python calculates. A person approves.




*Agent pattern today: one well-tooled dispatcher, run as a workflow. The model never picks a tool or does arithmetic.*

PRIMARY OBJECTIVE

# We declare: minimise lateness

## WHY

Quick response sells speed. A defect can be reworked; a missed reorder date loses the sale.

## MEASURED BY

The share of batches that arrive late, and the average days late, from the course simulator.

## BAR TO CLEAR

Beat random, biggest-capacity and cheapest on both measures, in normal runs and in the closure test.

## ALL FOUR COURSE OBJECTIVES: WINS AND LOSSES

| OBJECTIVE | OUR BEST SETTING | VERDICT |
|---|---|---|
| Minimise lateness | 1.7% late, vs 34.2% for random | **Beats all three baselines** |
| Minimise defects | 5.5% defective pieces, 18.4% late | **Biggest-capacity still wins (5.3%)** |
| Robust to failure | Closure adds 0.25 points late; random adds 4.0 | **Steadier than random and biggest. Cheapest stays 92% late either way** |
| Fair allocation | Not tuned yet: the busiest workshop takes 26.2% of pieces | **Random spreads work more evenly (22.6%)** |

*Course simulator, mean of 10 runs (seeds 5105-5114). The defects setting was chosen on separate development seeds before this run.*

# 95% fewer late batches than random allocation

**1.7%**

of batches late, 2 in 120, when queue, processing and transport are checked before choosing

**34.2%**

41 in 120 for a random eligible workshop

*Course simulator, 10 runs. Not a factory result.*

1 **Operational efficiency & risk reduction**

No bounced batches: the course's own eligibility rule runs before anything is recommended.

2 **Transparency & auditability**

A full record of the request, every option, the estimates, who set each constraint and who approved.

3 **Proactive exception management**

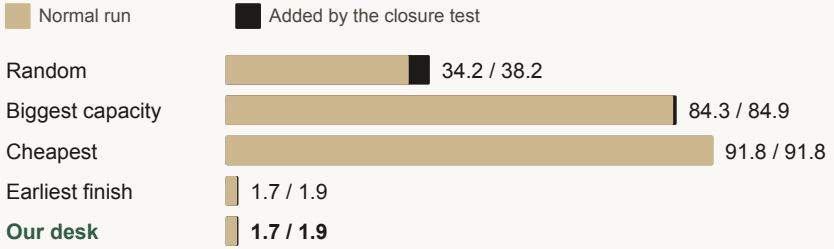
An impossible date gets the earliest achievable date and the options: split, accept late, escalate.

4 **Transparent cost-speed trade-offs**

$115.6k per run, against $86.0k for always-cheapest (late on 92% of batches) and $111.8k for random.

# Working allocator, working deployment pipeline

% OF BATCHES LATE, MEAN OF 10 RUNS. LOWER IS BETTER

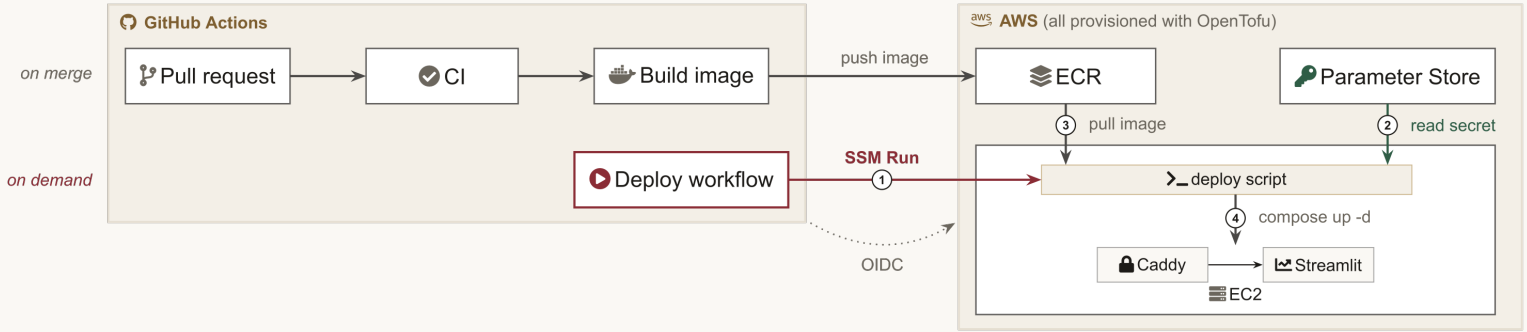



**HONEST FINDING**

Earliest finish ties us. Lateness is near its floor: two batches of 2,000 accessories cannot be made in their window.

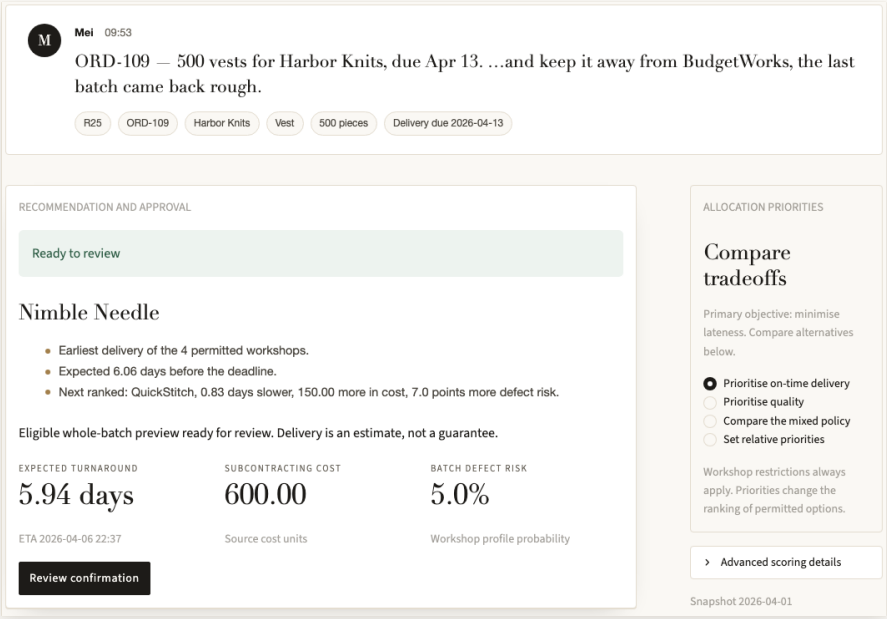
**WHAT OUR ALLOCATOR ADDS**

The same code, re-weighted: a 10-day buffer cuts cost 7.7%, below random's, and late days 23%, for about one more late batch per ten runs.




*Course simulator, seeds 5105-5114; buffer chosen on seeds 6000-6009 first. Sources: docs/V3_EVALUATION.md, docs/MARGIN_STUDY.md, docs/DEPLOYMENT.md (main).*

# Candidates, estimates and the choice, at a glance




1 **The message stays in view**

Request R25: "keep it away from BudgetWorks". The exclusion is applied and recorded against Mei, who asked.

2 **It says why, in Python's words**

Earliest of the permitted workshops, days to spare, and how the next option compares. No model writes these lines.

3 **Where the days go**

A chart splits each option into queue, processing, transport and rework, so a busy "fast" workshop shows up.

4 **Review before anything is saved**

Confirming rechecks the queues first, then writes one audit record.

**ALSO WORKING** All 30 inbox requests reach the right outcome from hand-checked inputs / live chat that asks when unsure / stale-safe saved plan / 300 offline tests

# What could go wrong, and what we assume

| RISK | CHANCE / IMPACT | WHAT WE DO ABOUT IT |
|---|---|---|
| The model misreads a request | **High** / **High** | Every field must quote the message; Python checks it and asks when unsure. First live run: one needless question blocked a refusal. |
| A message tries to bend the rules | **Medium** / **High** | Eligibility is a Python check the model cannot switch off. A suspended workshop is refused; exclusions cannot be relaxed mid-chat. |
| A number no tool produced | **Medium** / **High** | Tools format every number, and an automatic check compares each figure on screen with tool output. |
| Tuning to the 120 supplied orders | **Medium** / **Medium** | Choose settings on separate development seeds, then run the course seeds once. |
| The simulator is simpler than a factory | **Certain** / **Medium** | Say so, keep its default run as the yardstick, and test the desk with dispatchers. |

## KEY ASSUMPTIONS

Each inbox request is judged on the 1 April queues, on its own.

A defect means half the batch is redone, as in the simulator.

Workshop prices are our subcontract cost, not the customer’s price, so price questions are declined.

A person confirms every allocation. No workshop is contacted automatically.

# From working tools to an AI-native desk

| | WORK | SUCCESS CRITERION |
|---|---|---|
| 1 | Evaluation set first | 30+ questions checked against the data, 10 held out, 5+ injection and bait items |
| 2 | Live chat on all 30 inbox requests | Labels two teammates write independently; decision and explanation scored apart |
| 3 | Tool calling for questions | "Who can take 1,500 crewnecks?": the model picks a read-only tool, Python answers |
| 4 | Grounded explanations and replies | The model writes with placeholders; Python fills every number, nothing auto-sends |
| 5 | LLM judge, checked against people | Agreement with both human labellers reported; people stay the ground truth |
| 6 | Freeze the second objective and hybrid | Weights fixed on development seeds, metrics on a common scale, one course run |
| 7 | Demo video | Request, clarify, refuse, impossible date, confirm |

**OUR SUCCESS CRITERION** Live chat gets the right behaviour on at least 27 of 30 inbox requests against independent labels, with no number on screen that lacks a tool source, while the desk still beats all three baselines.

SWEATERCO

# Thank you.

*Questions are welcome.*

THE MODEL READS / PYTHON CALCULATES / A PERSON APPROVES